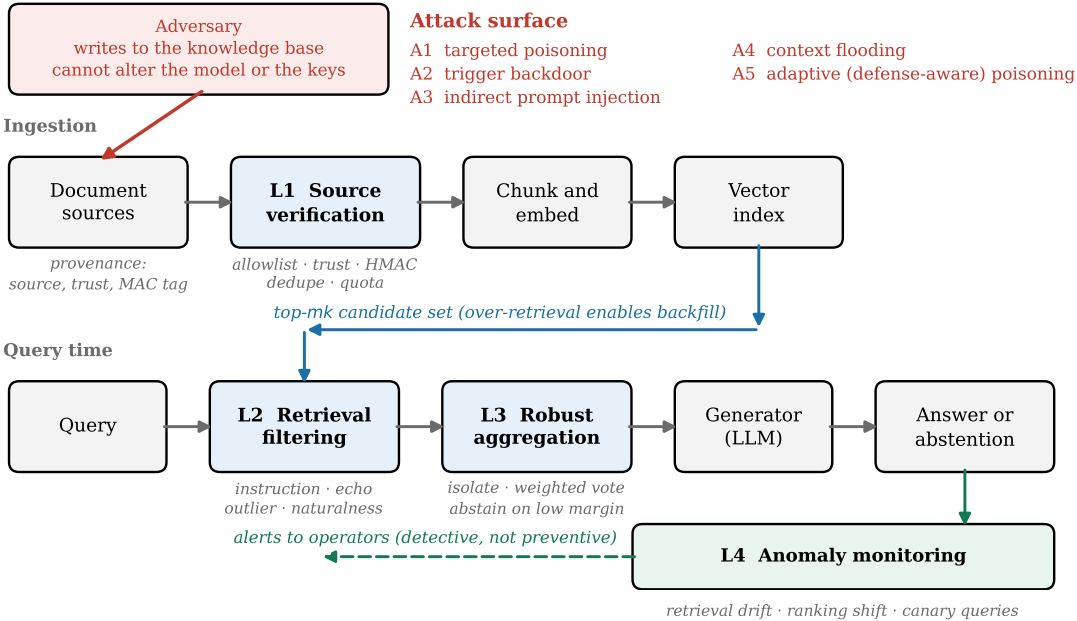
Adversary
writes to the knowledge base
cannot alter the model or the keys
Attack surface
A1 targeted poisoning
A2 trigger backdoor
A3 indirect prompt injection
A4 context flooding
A5 adaptive (defense-aware) poisoning
Ingestion
Document sources
L1 Source verification
Chunk and embed
Vector index
provenance: source, trust, MAC tag
allowlist · trust · HMAC dedupe · quota
top-mk candidate set (over-retrieval enables backfill)
Query time
Query
L2 Retrieval filtering
L3 Robust aggregation
Generator (LLM)
Answer or abstention
instruction · echo outlier · naturalness
isolate · weighted vote abstain on low margin
alerts to operators (detective, not preventive)
L4 Anomaly monitoring
retrieval drift · ranking shift · canary queries